

Lemma 3. Let V be a vector space (over F), if w is a linear combination of vectors $v_1, v_2, \dots, v_k$ and each v_i for $1 \leq i \leq k$ is a linear combination of vectors $u_1, u_2, \dots, u_l$, then w is a linear combination of vectors $u_1, u_2, \dots, u_l$.

Proof. By assumption we have that $w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k$ with $\lambda_1, \lambda_2, \dots, \lambda_k \in F$ and for each $1 \leq i \leq k$ $v_i = \lambda_{i1} u_1 + \lambda_{i2} u_2 + \dots + \lambda_{il} u_l$. Now (after substitution and grouping terms) one gets $w = \lambda_1(\lambda_{11} u_1 + \lambda_{12} u_2 + \dots + \lambda_{1l} u_l) + \lambda_2(\lambda_{21} u_1 + \lambda_{22} u_2 + \dots + \lambda_{2l} u_l) + \dots + \lambda_k(\lambda_{k1} u_1 + \lambda_{k2} u_2 + \dots + \lambda_{kl} u_l) = (\lambda_1 \lambda_{11} + \lambda_2 \lambda_{21} + \dots + \lambda_k \lambda_{k1}) u_1 + (\lambda_1 \lambda_{12} + \lambda_2 \lambda_{22} + \dots + \lambda_k \lambda_{k2}) u_2 + \dots + (\lambda_1 \lambda_{1l} + \lambda_2 \lambda_{2l} + \dots + \lambda_k \lambda_{kl}) u_l$ and so w is a linear combination of $u_1, u_2, \dots, u_l$. $\square$

Some properties of ‘span’

Proposition 15. Let V be a vector space (over F) and A, B subsets of V , then

- (1) $A \subseteq \text{Span}(A)$;
- (2) If $A \subseteq B$, then $\text{Span}(A) \subseteq \text{Span}(B)$;
- (3) $\text{Span}(\text{Span}(A)) = \text{Span}(A)$

Proof. (1) and (2) are easy.

(3) From (1) we have that $\text{Span}(A) \subseteq \text{Span}(\text{Span}(A))$, now if $w \in \text{Span}(\text{Span}(A))$ we have that $w \in \text{Span}(A)$ using previous lemma. $\square$

Linear dependence and linear independence

Definition 33. Let V be a vector space (over F), then

- (1) Vectors $v_1, v_2, \dots, v_k$ are said to be linearly dependant if there exists scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ with at least one of them non-zero such that $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k = \mathbf{0}$.
- (2) Vectors $v_1, v_2, \dots, v_k$ are said to be linearly independent if whenever $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k = \mathbf{0}$, we have $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$

Examples: (1) Consider $V = \mathbb{R}^3$ with usual addition and scalar multiplication

- Vectors $(1, 2, 5), (0, 3, 1), (1, 5, 6)$ are linearly dependant since we have $(1, 5, 6) - (1, 2, 5) - (0, 3, 1) = (0, 0, 0)$.
- Vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ are linearly independent for if we have $\lambda_1(1, 0, 0) + \lambda_2(0, 1, 0) + \lambda_3(0, 0, 1) = (0, 0, 0)$, then $(\lambda_1, \lambda_2, \lambda_3) = (0, 0, 0)$, and so $\lambda_1 = \lambda_2 = \lambda_3 = 0$.

(2) Let P_3 denote the set of all polynomials with degree of at most 3, are polynomials $p_1 = 1 + x^2$, $p_2 = 1 - x^2 + 2x^3$ and $p_3 = 1 - x$ linearly dependant?

This is equivalent to asking if there exists scalars $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ with at least one scalar non-zero such that $\lambda_1 p_1 + \lambda_2 p_2 + \lambda_3 p_3 = 0$, that is, $\lambda_1(1 + x^2) + \lambda_2(1 - x^2 + 2x^3) + \lambda_3(1 - x) = 0$, equivalently,

$(\lambda_1 + \lambda_2 + \lambda_3) - \lambda_3 x + (\lambda_1 - \lambda_2)x^2 + 2\lambda_2 x^3 = 0$. We then have the following linear system

$$\begin{aligned}\lambda_1 + \lambda_2 + \lambda_3 &= 0 \\ -\lambda_3 &= 0 \\ \lambda_1 - \lambda_2 &= 0 \\ 2\lambda_2 &= 0\end{aligned}$$

from which we get $\lambda_1 = \lambda_2 = \lambda_3 = 0$ and so p_1, p_2 and p_3 are not linearly dependant.

Theorem 14. *Let V be a vector space (over F), then*

- (1) *Vectors $v_1, v_2, \dots, v_k$ are linearly dependant if and only if at least one of these vectors can be written as a linear combination of the other vectors.*
- (2) *Vectors $v_1, v_2, \dots, v_k$ are linearly independent if and only if none of each of the these vectors is a linear combination of the other vectors.*

Proof. (1) Suppose $v_1, v_2, \dots, v_k$ are linearly dependant, then there exists $\lambda_1, \lambda_2, \dots, \lambda_k$ such that at least one λ_i (with $1 \leq i \leq k$) is non-zero, suppose without loss of generality that $\lambda_1 \neq 0$, then from $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k = \mathbf{0}$ we get $v_1 = -\frac{\lambda_2}{\lambda_1} v_2 - \frac{\lambda_3}{\lambda_1} v_3 - \dots - \frac{\lambda_k}{\lambda_1} v_k$ and so v_1 is a linear combination of vectors $v_2, \dots, v_k$, conversely if we suppose (without loss of generality) that v_1 is a linear combination of $v_2, \dots, v_k$ then we have that $v_1 = \lambda_2 v_2 + \lambda_3 v_3 + \dots + \lambda_k v_k$ and so $-v_1 + \lambda_2 v_2 + \lambda_3 v_3 + \dots + \lambda_k v_k = \mathbf{0}$, notice that $-v_1$ has a non-zero scalar and so vectors $v_1, v_2, \dots, v_k$ are linearly dependant.

(2) Check it! □

In $\mathbb{R}^n$ with usual addition and scalar multiplication.

Lemma 4. *A homogeneous system of n linear equations with k unknowns always has a non-trivial solution if $k > n$.*

Proof. The linear system being homogeneous it has at least one solution, and thus consistent. Now, since it has n equations, its associated augmented matrix in row echelon form will have at most n pivots and at least $k - n$ free variables since $k > n$ and so we will always have a non-trivial solution. □

Theorem 15. *If $v_1, v_2, \dots, v_k$ are vectors in $\mathbb{R}^n$ and $k > n$, then vectors $v_1, v_2, \dots, v_k$ are linearly dependant.*

Proof. Let $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$ and

$$\begin{aligned}\mathbf{v}_1 &= (a_{11}, a_{12}, \dots, a_{1n}) \\ \mathbf{v}_2 &= (a_{21}, a_{22}, \dots, a_{2n}) \\ &\vdots \\ \mathbf{v}_k &= (a_{k1}, a_{k2}, \dots, a_{kn})\end{aligned}$$

then from the first equation we get linear system

$$\begin{aligned}a_{11}\lambda_1 + a_{21}\lambda_2 + \cdots + a_{k1}\lambda_k &= 0 \\a_{21}\lambda_1 + a_{22}\lambda_2 + \cdots + a_{k2}\lambda_k &= 0 \\&\vdots \\a_{1n}\lambda_1 + a_{2n}\lambda_2 + \cdots + a_{kn}\lambda_k &= 0\end{aligned}$$

but since $k > n$, using the previous lemma we have that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly dependant. $\square$

Example: In $\mathbb{R}^3$ with usual addition and scalar multiplication, vectors $(\pi, 1, 2)$, $(4, 5, 100)$, $(2070, 1, 0)$ and $(10, 1, 3)$ are linearly dependent since they are 4 vectors in $\mathbb{R}^3$.